Amersham Cy3 Maleimide Mono-Reactive Dye 5-pack

Product Specification Sheet

Introduction

Product code

PA23031

About

Reagents for the labeling of biological compounds with Cy™3 monofunctional dye.

Important

Read these instructions carefully before using the products.

Intended use

The products are intended for research use only, and shall not be used in any clinical or *in vitro* procedures for diagnostic purposes.

Safety

For use and handling of the products in a safe way, refer to the Safety Data Sheets.

CAUTION

This dye is intensely colored and very reactive.

Care must be exercised when handling the dye vial to avoid staining clothing, skin, and other items.

Storage

Store refrigerated at 2–8°C in the dark. Do not use if desiccant capsule in foil pack is either pink or green.

Expiry

See outer packaging.

Components

- Five foil packs: each containing dried dye to label 1 mg of protein.
- Product specification sheet: with instructions for using the dye.

Other materials required

- Conjugation buffer: 10–100 mM Phosphate (such as Phosphate-Buffered Saline (PBS)), Tris or HEPES buffer with pH between 7.0–7.5.
- Dimethylformamide, anhydrous (DMF): for preparing dye solutions.
- Tris-(2-Carboxethyl) Phosphine (TCEP): for reducing Disulphide bonds.
- Separation column: containing a permeation gel (such as Bio-Gel™ P-6 gel, minimum of 10 mL bed volume and 6 cm packed length).
- Separation buffer: Phosphate-Buffered Saline, pH 7–7.5, containing 0.1% Sodium Azide.
- Test tubes.
- Transfer pipettes.
- Glassware.

Background

Cyanine reagents have been shown to be useful as fluorescent labels for biological compounds (1, 2).

Cy3 dye is an orange fluorescing cyanine that produces an intense signal easily detected using fluorescein filter sets. The Cy3 dye supplied here is a mono-functional Maleimide ready for the labeling of compounds containing free sulphydryl groups (3).

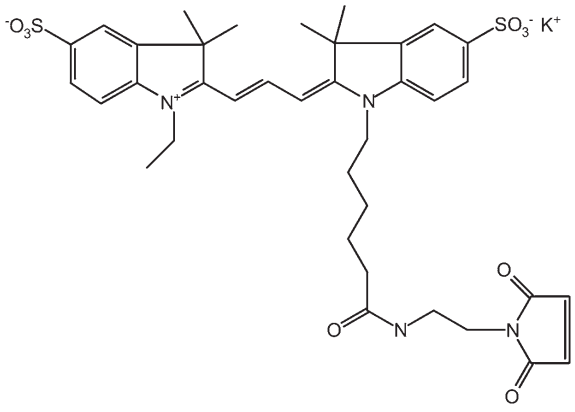


Fig 1. Cy3 monofunctional dye.

Recommended procedure for use

This protocol has been designed for the preparation of Cy3-labeled IgG antibodies. It is designed to label 1 mg protein to a final molar dye/protein (D/P) ratio between 0.5 and 3. This assumes an average protein molecular weight of 155 000 daltons.

Note: *The following materials and procedures are used in a functional test of the dye to label reduced IgG antibodies. Other proteins may also be readily labeled, however, choice of buffers, separation media, and technique may vary in order to produce optimal results. TCEP is used to reduce the IgG. Other reducing reagents such as 2-Mercaptoethylamine Hydrochloride may give a more selective reduction of disulphide bonds (4).*

Altering the protein concentration and reaction pH will change the labeling efficiency of the reaction. The optimal pH for the reaction of Maleimides is near 7.0. In the pH range 7.0–7.5 the protein thiol groups are sufficiently nucleophilic so that they almost exclusively react with the Maleimide dye in the presence of the more numerous protein amines, which are protonated and relatively unreactive.

Conjugation of dye to antibody

To minimize oxidation of thiols, carry out thiol modifications in an oxygen-free environment, that is under Nitrogen, using degassed solvents / buffers.

Antibody to be conjugated must be dissolved at 1 mg/mL in degassed PBS buffer, being careful not to introduce air bubbles into the solution.

Step	Action
1	Leave the solution for 30 minutes at room temperature.
2	Add a 100 molar excess of TCEP (180 µg, 10 µL of a 18 mg/mL TCEP solution in PBS, per 1 mg of IgG).
3	Flush the vial with nitrogen gas, cap the vial, and mix thoroughly.
4	Incubate the reaction at room temperature for 10 minutes. It is not necessary to remove excess TCEP before conjugation.
5	While the IgG reduction is taking place prepare a dye solution by adding 50 µL of anhydrous Dimethylformamide to one pack of dye.
6	Flush the vial with Nitrogen gas, cap the vial, and mix thoroughly.
7	Add the dye solution (50 µL) to 1010 µL of reduced IgG.
8	Flush the vial with Nitrogen gas, cap the vial and mix thoroughly.
9	Incubate the reaction at room temperature for two hours with additional mixing every 30 minutes. Then leave the reaction overnight at 2–8°C.

Separation of protein from free dye

Labeled antibody can be separated from the excess, unconjugated dye by gel permeation chromatography. It is convenient to pre-equilibrate the column with Phosphate-Buffered Saline and to elute the protein using the same buffer. Two pink bands should develop during elution. The faster moving band is Cy3-labeled antibody while the slower band is free dye. For precise separation 1 mL fractions should be collected, analyzed and the desired fractions pooled. Many Cy3-labeled proteins can be stored at 2–8°C without further manipulation.

Labeled antibody can also be separated from unconjugated dye by dialysis. Dialysis does not give as efficient and rapid a separation as gel filtration. We therefore recommend that protein purification by gel filtration be used.

Estimation of final dye/protein (D/P) ratio

Step	Action
1	Dilute a portion of the labeled protein solution so that the maximum absorbance is 0.5 to 1.5 AU. Molar concentrations of dye and protein are calculated, and the ratio of these values is the average number of dye molecules coupled to each protein molecule.

Molar extinction coefficients of 150 000 M^{-1} cm^{-1} at 552 nm for the Cy3 dye and 170 000 M^{-1} cm^{-1} at 280 nm for the protein are used in this example.

The extinction coefficient will vary for different proteins. The calculation is corrected for the absorbance of the dye at 280 nm (approximately 8% of the absorbance at 552 nm).

$[\text{Cy3 dye}]=(A_{552}) / 150\,000$

$[\text{antibody}]=[A_{280} - (0.08 \bullet A_{552})] / 170\,000$

$(D/P)_{final}=[\text{dye}] / [\text{antibody}]$

$(D/P)_{final}=[1.13 \bullet (A_{552})] / [A_{280} - (0.08 \bullet A_{552})]$

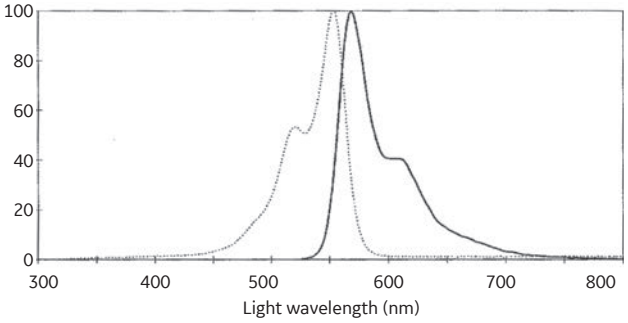


Fig 2. Cy3 dye absorption and fluorescence spectra.

Cy3 monofunctional dye characteristics

Formula weight	791.0
Absorbance max	550 nm
Extinction max	150 000 M^{-1} cm^{-1}
Emission max	570 nm

References

1. MUJUMDAR, R.B. *et al., Bioconjugate Chemistry* **4** (2), pp.105-111, 1993.
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3. NEBLOCK, D.S. *et.al., Bioconjugate Chemistry* **3** (2), pp.126-131, (1992).
4. ASLAM, M., and DENT, A., *Bioconjugation*, Macmillan Reference Ltd, (1998).

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28954681 AF V:7 02/2021

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